Lab 8.2: Database CRUD (SQLite and MongoDB) in F#

lab8.2.fsx

```
#r "nuget: Microsoft.Data.Sqlite, 8.0.0"
#r "nuget: MongoDB.Driver, 2.28.0"

open System
open Microsoft.Data.Sqlite
open MongoDB.Bson
open MongoDB.Driver

let execute (conn: SqliteConnection) (sql: string) (args: (string * obj) list) =
    use cmd = conn.CreateCommand()
    cmd.CommandText <- sql
    for (name, value) in args do
        cmd.Parameters.AddWithValue(name, value) |> ignore
    cmd.ExecuteNonQuery() |> ignore

let showStudents (conn: SqliteConnection) =
    use cmd = conn.CreateCommand()
    cmd.CommandText <- "SELECT Id, Name, Course, Marks FROM Students ORDER BY Id"
    use reader = cmd.ExecuteReader()
    let mutable any = false
    while reader.Read() do
        any <- true
        printfn "  Id=%d | %s | %s | %.1f" (reader.GetInt32(0)) (reader.GetString(1)) (reader.GetString(2)) (reader.GetDouble(3))
    if not any then printfn "  (no records)"

let runSqlite () =
    printfn "=========== SQLite (relational) ==========="
    use conn = new SqliteConnection("Data Source=students.db")
    conn.Open()
    execute conn "DROP TABLE IF EXISTS Students" []
    execute conn "CREATE TABLE Students (Id INTEGER PRIMARY KEY AUTOINCREMENT, Name TEXT, Course TEXT, Marks REAL)" []

    printfn "CREATE: insert two students"
    execute conn "INSERT INTO Students (Name, Course, Marks) VALUES (@n, @c, @m)" [ "@n", box "Asha"; "@c", box "CS"; "@m", box 85.5 ]
    execute conn "INSERT INTO Students (Name, Course, Marks) VALUES (@n, @c, @m)" [ "@n", box "Rohan"; "@c", box "IT"; "@m", box 72.0 ]
    showStudents conn

    printfn "UPDATE: change Rohan's marks to 90"
    execute conn "UPDATE Students SET Marks = @m WHERE Name = @n" [ "@m", box 90.0; "@n", box "Rohan" ]
    showStudents conn

    printfn "DELETE: remove Asha"
    execute conn "DELETE FROM Students WHERE Name = @n" [ "@n", box "Asha" ]
    showStudents conn

let makeDoc (name: string) (course: string) (marks: float) =
    BsonDocument("name", BsonString name).Add("course", BsonString course).Add("marks", BsonDouble marks)

let showDocs (collection: IMongoCollection<BsonDocument>) =
    let docs = collection.Find(Builders<BsonDocument>.Filter.Empty).ToList()
    if docs.Count = 0 then printfn "  (no records)"
    for d in docs do
        printfn "  %s | %s | %.1f" d.["name"].AsString d.["course"].AsString d.["marks"].AsDouble

let runMongo () =
    printfn "\n=========== MongoDB (NoSQL) ==========="
    let settings = MongoClientSettings.FromConnectionString("mongodb://localhost:27017")
    settings.ServerSelectionTimeout <- TimeSpan.FromSeconds 5.0
    let client = MongoClient(settings)
    let collection = client.GetDatabase("college").GetCollection<BsonDocument>("students")
    collection.DeleteMany(Builders<BsonDocument>.Filter.Empty) |> ignore
```

Lab 8.2: Database CRUD (SQLite and MongoDB) in F# (contd.)

```
62
63      printfn "CREATE: insert two students"
64      collection.InsertOne(makeDoc "Asha" "CS" 85.5)
65      collection.InsertOne(makeDoc "Rohan" "IT" 72.0)
66      showDocs collection
67
68      printfn "UPDATE: change Rohan's marks to 90"
69      collection.UpdateOne(Builders<BsonDocument>.Filter.Eq("name", "Rohan"), Builders<BsonDocument>.Update.Se
    t("marks", 90.0)) |> ignore
70      showDocs collection
71
72      printfn "DELETE: remove Asha"
73      collection.DeleteOne(Builders<BsonDocument>.Filter.Eq("name", "Asha")) |> ignore
74      showDocs collection
75
76  runSqlite ()
77
78  try
79      runMongo ()
80  with ex ->
81      printfn "MongoDB error: %s" ex.Message
82      printfn "Make sure MongoDB is running on localhost:27017."
```